LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 5

Course Code: CSE252
Course Title: INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
Course Outcome: CO5
Paper: Sample Multiple Choice Questions - ANSWER KEY
Maximum Marks: 40 x 1 = 40

Unit: 5
Unit Title: Model Evaluation and Improvement
Unit Mapped RBT: L3 (Apply)
Total Questions: 40
Suggested Duration: 60 Minutes

Answer Grid (Key)

Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
Q1	A	Q11	D	Q21	A	Q31	C
Q2	B	Q12	B	Q22	C	Q32	A
Q3	C	Q13	D	Q23	B	Q33	A
Q4	D	Q14	C	Q24	D	Q34	D
Q5	B	Q15	B	Q25	C	Q35	B
Q6	D	Q16	A	Q26	D	Q36	C
Q7	A	Q17	B	Q27	A	Q37	C
Q8	C	Q18	A	Q28	B	Q38	B
Q9	C	Q19	D	Q29	D	Q39	D
Q10	A	Q20	C	Q30	B	Q40	A

Multiple Choice Questions (40 x 1 = 40 Marks)

Q1. Name the regression metric that averages the absolute differences between the predictions and the actual values.

A) Mean absolute error <-- correct
B) Mean squared error
C) Root mean squared error
D) R squared

Answer: A. Mean absolute error treats every unit of error alike, which makes it easier to read and less sensitive to a few large mistakes.

Q2. Identify the entry of a confusion matrix that counts the positive cases correctly identified.

A) False positives
B) True positives <-- correct
C) True negatives
D) False negatives

Answer: B. The four entries of the matrix are the basis of every classification metric, and reading them correctly is the first step in any evaluation.

Q3. State what the F1 score combines.

A) Accuracy and recall, as their arithmetic mean
B) Precision and accuracy, as their product
C) Precision and recall, as their harmonic mean <-- correct
D) Recall and specificity, as their difference

Answer: C. The harmonic mean is low unless both quantities are reasonably high, so F1 cannot be inflated by one of the two alone.

Q4. Name the regression metric obtained by taking the square root of the mean squared error.

A) Mean absolute error
B) R squared
C) Adjusted R squared
D) Root mean squared error <-- correct

Answer: D. Taking the root returns the figure to the units of the target, which is why it is reported more often than the mean squared error itself.

Q5. Identify the two quantities plotted against one another on a ROC curve.

A) Precision against recall
B) True positive rate against false positive rate <-- correct
C) Accuracy against the threshold
D) Training error against test error

Answer: B. The curve is traced by sweeping the decision threshold, and the area beneath it summarises the classifier over every threshold at once.

Q6. State what a global explanation of a model describes.

A) Why one particular record received its prediction

B) How long the model takes to make a prediction
C) How much memory the model needs to run
D) How the model behaves across the whole population <-- correct
Answer: D. A global explanation names the features that matter overall. A person affected by one decision needs the local explanation instead.

Q7. Explain why the root mean squared error is expressed in the same units as the target.

A) The square root undoes the squaring of the errors <-- correct
B) The errors are divided by the number of records
C) The metric is always scaled between zero and one
D) The metric ignores the sign of every error
Answer: A. A mean squared error on a price in rupees is in rupees squared, which no one can interpret. The root restores a readable figure.

Q8. Distinguish precision from recall by the quantity each is measured over.

A) Recall is over the predicted positives, precision over the actual
B) Both of the two are measured over the predicted positives
C) Precision is over the predicted positives, recall over the actual positives <-- correct
D) Both of the two are measured over the actual positives
Answer: C. Precision asks how many alarms were genuine; recall asks how many genuine cases were caught. The two answer different questions.

Q9. Describe what an R squared of exactly zero indicates about a regression model.

A) It predicts every value perfectly correctly
B) It has an average error of exactly zero units
C) It does no better than always predicting the mean <-- correct
D) It cannot be applied to this kind of data
Answer: C. R squared compares the model's residual error with the error of the simplest possible predictor. A value of zero means it has added nothing.

Q10. Interpret a ROC-AUC of 0.5 reported for a classifier.

A) It performs no better than random guessing <-- correct
B) It performs perfectly on every record
C) It has an accuracy of exactly 50 per cent
D) It always predicts the positive class
Answer: A. An area of 1.0 is a perfect ranking and 0.5 is the diagonal, which is what a coin toss would achieve.

Q11. Predict the effect on precision of raising the decision threshold from 0.5 to 0.8.

A) Precision usually falls while recall rises
B) Both precision and recall usually rise
C) Neither precision nor recall is affected
D) Precision usually rises while recall falls <-- correct
Answer: D. A higher bar means fewer records are called positive, so those that are tend to be more certain, but genuine cases are missed.

Q12. Recognize the situation described: a fraud model raises very few false alarms but catches only a small share of the frauds.

A) High recall with low precision
B) High precision with low recall <-- correct
C) High values of both of the two
D) Low values of both of the two
Answer: B. Whether this is acceptable depends entirely on what a missed fraud costs compared with the cost of investigating a false alarm.

Q13. Explain why the mean squared error penalises a large error more heavily than the mean absolute error does.

A) Squaring removes the sign of the error
B) Squaring divides the error by the record count
C) Squaring converts the error into a percentage
D) Squaring grows faster than the error itself <-- correct
Answer: D. An error of 10 contributes 100 to the squared total but only 10 to the absolute one, so a handful of large mistakes dominate the mean squared error.

Q14. Summarize what cross validation gives that a single train-test split does not.

A) A model that is trained on far more data
B) A guarantee that the model will not overfit
C) An estimate that does not depend on one lucky split <-- correct
D) A final performance figure for the report
Answer: C. Every row serves in turn for validation, so the estimate is averaged over several partitions and its variability can also be reported.

Q15. Distinguish grid search from random search by the way each chooses the values it tries.

A) Random tries every combination, grid samples them
B) Grid tries every combination, random samples them <-- correct
C) Both of the two try every combination in turn
D) Both of the two sample the combinations at random
Answer: B. With many hyperparameters the grid becomes impossibly large, and random sampling often finds a good setting for a fraction of the work.

Q16. Explain why explainability matters to an applicant a model has rejected.

A) They cannot challenge a decision they cannot understand <-- correct

B) They would otherwise be unable to reapply

C) The explanation makes the model more accurate

D) The explanation shortens the processing time

Answer: A. The person needs the local explanation for their own case; the average behaviour of the model tells them nothing about why they were refused.

Q17. Predict what happens when the hyperparameters are tuned by repeatedly checking the test set.

A) The model becomes impossible to train

B) The test score stops being an honest estimate <-- correct

C) The test set grows larger with each check

D) The hyperparameters cease to have any effect

Answer: B. Each choice made on the strength of the test score leaks that set into the model. A separate validation partition is what keeps the test set clean.

Q18. Describe what a local explanation accounts for.

A) Why one particular record received the prediction it did <-- correct

B) Which features matter across all the records

C) How long the model took to be trained

D) How many records were used in training

Answer: A. A local explanation is what a person affected by an automated decision is entitled to, since the global picture does not address their case.

Q19. Explain why a single accuracy figure can hide poor performance on one class.

A) Accuracy is computed on the training data only

B) Accuracy cannot be computed for two classes

C) Accuracy always equals the recall of the model

D) A small class contributes little to the overall figure <-- correct

Answer: D. A class holding two per cent of the records can be handled badly while the headline figure stays high, which is why results are broken down by class.

Q20. Distinguish a false positive from a false negative in a medical screening test.

A) A false negative alarms a healthy person

B) Both of the two miss an ill person

C) A false positive alarms a healthy person, a false negative misses an ill one <-- correct

D) Both of the two alarm a healthy person

Answer: C. The two errors carry very different costs here, which is why recall is usually weighted more heavily than precision in screening.

Q21. Compute the recall of a model that records 30 true positives and 20 false negatives.

A) 0.6 <-- correct

B) 0.4

C) 0.75

D) 1.5

Answer: A. Recall is true positives divided by all actual positives: 30 / (30 + 20) = 30 / 50 = 0.6.

Q22. Compute the precision of a model that records 30 true positives and 10 false positives.

A) 0.25

B) 0.6

C) 0.75 <-- correct

D) 3.0

Answer: C. Precision is true positives divided by all predicted positives: 30 / (30 + 10) = 30 / 40 = 0.75.

Q23. Compute the accuracy of a model recording 40 true positives, 50 true negatives, 5 false positives and 5 false negatives.

A) 0.8

B) 0.9 <-- correct

C) 0.45

D) 0.95

Answer: B. Accuracy is the correct predictions over the total: (40 + 50) / 100 = 0.9.

Q24. Solve the following: a model records a precision of 0.6 and a recall of 0.4. Determine its F1 score.

A) 0.50

B) 0.24

C) 1.00

D) 0.48 <-- correct

Answer: D. F1 is 2 x (0.6 x 0.4) / (0.6 + 0.4) = 2 x 0.24 / 1.0 = 0.48.

Q25. Compute the mean absolute error of a model whose three errors are 2, 3 and 5.

A) 10

B) 5

C) About 3.33 <-- correct

D) About 3.87

Answer: C. The mean absolute error is (2 + 3 + 5) / 3 = 10 / 3, which is approximately 3.33.

Q26. Compute the mean squared error of a model whose two errors are 1 and 3.

A) 2

B) 4

C) 10

D) 5 <-- correct

Answer: D. The mean squared error is (1 squared + 3 squared) / 2 = (1 + 9) / 2 = 5.

Q27. Solve the following: a model records a mean squared error of 49. Determine its root mean squared error.

A) 7 <-- correct
B) 24.5
C) 49
D) 2401
Answer: A. The root mean squared error is the square root of 49, which is 7, expressed in the units of the target.

Q28. Apply 4-fold cross validation to a dataset of 800 rows and determine the rows placed in the validation partition of each fold.

A) 600
B) 200 <-- correct
C) 400
D) 4
Answer: B. The data is divided into four equal folds of 800 / 4 = 200 rows, each serving once as the validation partition.

Q29. Compare a grid search over 3 values of one setting and 4 of another with a random search of 6 trials, on the number of combinations each evaluates.

A) The grid evaluates 7, the random search 6
B) The grid evaluates 6, the random search 12
C) Both of the two evaluate exactly 12
D) The grid evaluates 12, the random search 6 <-- correct
Answer: D. A grid evaluates the product of the value counts, 3 x 4 = 12, while a random search evaluates only the number of trials it is given.

Q30. Examine a classifier recording 99 per cent accuracy on a dataset in which 99 per cent of the records are negative, and determine the likely explanation.

A) It has learned the positive class extremely well
B) It predicts the negative class for every record <-- correct
C) It has been trained for too many iterations
D) The accuracy has been computed incorrectly
Answer: B. Always answering with the majority class achieves exactly this figure. Precision, recall and the confusion matrix expose the failure at once.

Q31. Use a change of the decision threshold from 0.5 to 0.7 and determine the effect on recall.

A) Recall rises
B) Recall is unchanged
C) Recall falls <-- correct
D) Recall becomes undefined
Answer: C. Fewer records clear the higher bar, so more genuine positives are missed. Precision usually moves in the opposite direction.

Q32. Apply the definition of R squared to a model whose residual sum of squares is 30 and whose total sum of squares is 150.

A) 0.8 <-- correct
B) 0.2
C) 5.0
D) 0.5
Answer: A. R squared is 1 - 30 / 150 = 1 - 0.2 = 0.8, so the model accounts for eighty per cent of the variation.

Q33. Apply 10-fold cross validation and determine how many times the model is fitted.

A) 10 <-- correct
B) 1
C) 9
D) 100
Answer: A. Each fold in turn is held out while the model is fitted on the remaining nine, so ten fits are performed.

Q34. Examine a model scoring 0.90 in cross validation but only 0.65 on data collected some months later, and determine the likely cause.

A) The cross validation used too many folds
B) The model was trained for too few iterations
C) The metric was computed with the wrong formula
D) The incoming data has drifted from the training data <-- correct
Answer: D. Both readings are honest; the population has changed. Monitoring detects this and retraining on recent data is the usual response.

Q35. Apply the confusion matrix to a screening test in which 100 cases are genuinely positive and 70 of them are detected. Determine the false negatives.

A) 70
B) 30 <-- correct
C) 100
D) 0
Answer: B. The positives that were missed are 100 - 70 = 30, and they are what recall measures.

Q36. Use ROC-AUC values of 0.85 and 0.62 to select between two candidate classifiers.

A) The model scoring 0.62
B) Neither, since both are unusable
C) The model scoring 0.85 <-- correct
D) Either, since the two are equivalent
Answer: C. A larger area means the model ranks positives above negatives more reliably across every threshold.

Q37. Apply a grid search over one setting with 3 values and another with 5, evaluated by 5-fold cross validation, and determine the number of model fits.

A) 15

B) 25

C) 75 <-- correct

D) 8

Answer: C. There are 3 x 5 = 15 combinations, and each is evaluated over 5 folds, giving 15 x 5 = 75 fits in total.

Q38. Examine a regression reporting a mean absolute error of 4 alongside a root mean squared error of 12, and determine what it indicates.

A) Every prediction is wrong by about the same amount

B) A few predictions are badly wrong <-- correct

C) The model predicts the mean for every record

D) The two metrics have been swapped by mistake

Answer: B. Squaring magnifies the large errors, so a root mean squared error far above the mean absolute error is the signature of a small number of severe mistakes.

Q39. Use the ideas of Explainable Artificial Intelligence to determine what a person refused a loan is entitled to.

A) A global explanation of the model

B) The full training dataset used

C) The source code of the model

D) A local explanation of their own case <-- correct

Answer: D. The average behaviour of the model across all applicants does not tell this applicant why their own application was refused.

Q40. Relate the number of hyperparameter combinations to the cost of running a grid search.

A) The cost is the product of the value counts times the folds <-- correct

B) The cost is the sum of the value counts

C) The cost does not depend on the number of folds

D) The cost falls as more settings are added

Answer: A. Because the grid grows multiplicatively, adding one more setting with five values multiplies the whole search by five, which is why random search is often preferred.

----- End of Sample Question Set -----